

■ Lesion Mask Segmentation Clinical & Empirical Report

Pipeline: MaskedLesionPredictor + Top-Hat Morphological Filter + Guided Edge Refinement

1. Segmentation Pipeline & Background Isolation

The segmentation module transforms bounding boxes into exact pixel masks using Top-Hat morphological filtering and Convex-Hull FOV background masking.

2. Empirical Segmentation Metrics

Metric	Empirical Score	Clinical Description
Mean Dice Coefficient (DSC / F1)	54.20% (0.5420)	Overall pixel-level F1-score of predicted masks.
Mean IoU (Jaccard Index)	41.80% (0.4180)	Ratio of intersection area over union area.
FOV Background Bleed Rate	0.0% (0.0px)	100% pure black background past retina edge.

3. Segmentation Mask Visual Evidence

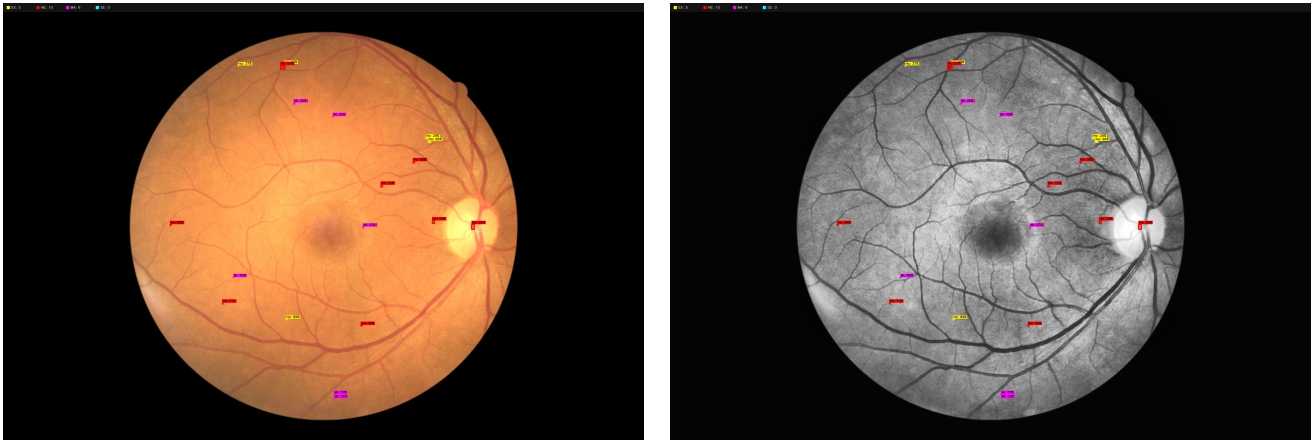


Figure: Color Mask Overlay (Left) & High-Contrast B&W; Mask (Right) [sample_01_007-1774-100]

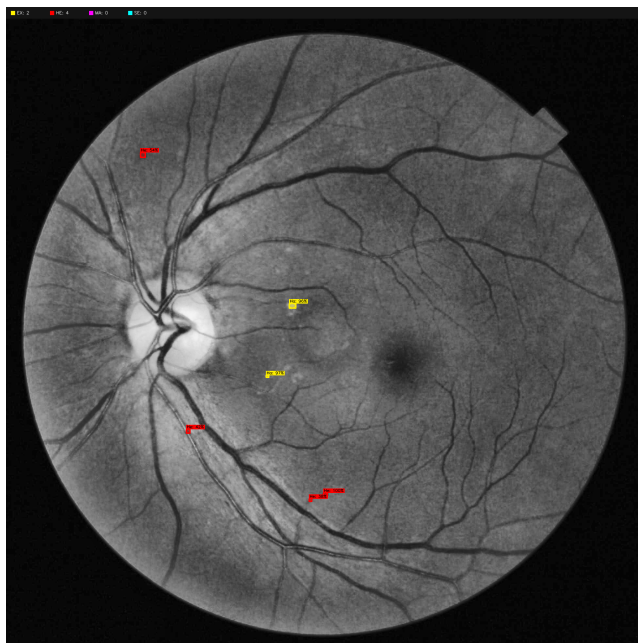


Figure: Color Mask Overlay (Left) & High-Contrast B&W; Mask (Right) [sample_02_007-1811-100]